

February 23, 2026

Response to the Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care (RIN 0955-AA13)

Thank you for the opportunity to provide comments on the *Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care*. As one of the nation's leading academic health systems, we appreciate the opportunity to provide comments as the administration considers opportunities to promote AI innovation and adoption in clinical care.

Duke Health integrates the Duke University Health System (all of our hospitals, clinics, and the Duke Health Integrated Practice), Duke University School of Medicine, Duke-NUS Medical School, Duke University School of Nursing, and incorporates the health and health research programs within the Duke Global Health Institute as well as those in schools and centers across Duke University, including the Duke Robert J. Margolis Institute for Health Policy.

Our organization is committed to excellence through the adoption of AI innovations to improve patient outcomes, reduce administrative burden, and streamline care delivery. To achieve this, we established the Algorithm-Based Clinical Decision Support (ABCDS) Oversight Committee in 2020. This committee governs and reviews AI solutions for patient care. At that time, AI quality and ethical considerations were not widely understood or adopted by stakeholders, and FDA guidance was still under development. Over the past five years, we have worked to balance innovation with high-quality, ethical care, ensuring responsible deployment of AI solutions.

Based on our experience, Duke Health offers the following considerations for promoting AI innovation and adoption in clinical care:

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Currently, significant variation in legislation across states creates barriers to widespread adoption. A shared federal legal and regulatory framework, developed in partnership with states, would help reduce inconsistencies and minimize the compliance burden on companies that would otherwise need to meet different requirements in each state.

HHS can play a critical role by establishing greater transparency requirements for AI solution testing, validation, and implementation practices. Building on the transparency elements recommended in HTI-1, additional guidance related to AI workflow integration and decision-making based on model outputs could serve as a practical playbook for healthcare organizations. Such transparency standards would help increase trust, provide operational clarity, and ultimately drive broader adoption of AI tools in healthcare settings.

Incentives are needed for both healthcare organizations and end users of AI solutions to adopt, purchase, and maintain these technologies. Strengthening infrastructure readiness will also be essential to support successful implementation. (See below)

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

AI has significant potential to improve health outcomes for all Americans. However, determining which AI solutions are truly effective remains a core challenge. Because AI models are fundamentally data-driven, their performance can vary substantially across health systems due to differences in patient populations, care delivery patterns, and data quality. Ongoing monitoring is therefore essential: only through real-world evaluation can organizations determine whether a given AI solution improves outcomes in their specific clinical environment.

Yet many healthcare organizations lack the necessary expertise to implement, validate, and monitor AI tools effectively. They may also lack the technical infrastructure required to deploy and maintain these algorithms at scale. Incentives that build organizational readiness for responsible AI adoption are critical. These could include grants from HRSA, ASTP/ONC, and the CMS Innovation Center to support AI readiness assessments, workforce development, and infrastructure modernization. Additionally, a national AI technical assistance program—modeled after ONC’s Health IT Regional Extension Centers—could provide hands-on support to organizations that currently lack in-house expertise.

Today, each health system evaluates AI solutions largely in isolation, determining for itself whether a tool is effective and whether its benefits outweigh its risks. A national approach would reduce this duplication of effort. National registries could facilitate information sharing across healthcare organizations by aggregating real-world evidence on AI model performance, clinical effectiveness, and safety risks. Such a system would accelerate learning, support continuous model improvement, and promote the adoption of AI tools with demonstrated benefit.

HHS could help lead this effort by establishing a federated national AI registry, supported by CMS and ASTP, to collect standardized, privacy-preserving performance metrics.¹ Participation could be required for AI tools used in federally funded care settings to ensure broad coverage and transparency. Duke Health, for example, is co-developing an information-sharing platform with Avanade called SAIGE that will enable organizations to share data on which AI solutions are deployed and which provide meaningful clinical value. This type of infrastructure could serve as a foundation for a national registry model.

The health care delivery system currently lacks appropriate incentive structures to support the adoption and sustained use of AI-based tools. Today, the return on investment for many AI solutions is driven primarily by staff time savings; return on health is not easily translated into return on investment, which, in turn, can make it difficult for organizations to justify investment. Payment models that recognize and reward the use of AI tools demonstrated to improve clinical quality, patient outcomes, or system efficiency would create a more durable incentive for adoption. Aligning reimbursement with outcomes would enable hospitals to invest in AI as a mechanism for improving care delivery rather than as a cost-cutting measure.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Looking ahead, the emergence of next-generation AI systems—particularly agentic AI—raises new legal and operational challenges. When healthcare professionals are partially or fully out of the decision-making loop, accountability for errors becomes ambiguous. Robust evaluation methods rooted in both automation safety and human-factors engineering will be essential to assess the outputs, behaviors, and failure modes of agentic AI. These approaches can help ensure that automated clinical actions remain predictable, effective, safe, and aligned with clinical standards, even as systems become more autonomous.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and postdeployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Because AI solutions can perform differently across healthcare organizations, both pre-deployment and post-deployment evaluation as well as ongoing monitoring should be supported and encouraged. To further promote the use of effective AI, HHS could consider

¹ [A Federated Registration System for Artificial Intelligence in Health - PubMed](#)

allowing validated AI interventions to contribute toward existing quality performance metrics. In addition, HHS could offer shared-savings bonuses for organizations that implement AI tools shown to reduce avoidable hospitalizations, decrease disparities, or measurably improve clinical outcomes.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

Alignment around technical standards and risk frameworks is essential to promote consistent, safe, and scalable adoption of AI in clinical care. Clear federal support for widely recognized technical and governance standards—such as structured AI lifecycle management, transparency requirements, and post-deployment monitoring practices—would help reduce duplication of effort across health systems and provide greater regulatory clarity for both developers and providers.

In parallel, targeted investment in workforce education and re-skilling is critical. Many healthcare organizations currently lack the clinical, technical, and operational expertise needed to implement, validate, and continuously monitor AI tools in real-world settings. Federal support for training programs, technical assistance, and AI-focused workforce development would increase trust among clinicians and administrators, strengthen organizational readiness, and ensure that AI tools are used appropriately, safely, and in ways that deliver measurable clinical value.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

Ambient listening solutions have reduced the amount of time clinicians spend on documentation, and patients report that these tools improve the quality of patient-provider interactions. But for many AI tools, it is too early to assign an “exceeded” or “fallen short” label on their implementation as health systems are in early stages of evaluating new tools. Examples include computer vision with AI models to aid in prevention of falls from a hospital bed and prevention of pressure ulcers. The same computer vision technology is likely to support workflow modification to improve efficiencies of acute care delivery, perioperative operations, Emergency Department operations, and the patient and staff experience in all these services.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

A key driver of successful AI adoption is ensuring that each AI solution addresses a clearly defined unmet clinical need. To achieve this, stakeholders across service lines and departments should be engaged early in the selection process, including the build–buy–partner decision. Governance bodies can play an essential coordinating role by guiding these decisions, establishing clear accountability for AI deployment, and promoting transparency around which AI solutions provide measurable benefit.

End-user champions can further support adoption by coordinating pilot projects, training peers, and educating potential users on the benefits of each AI tool as well as strategies for managing associated risks. To cultivate a culture of innovation and support AI scaling, organizations can designate AI ambassadors embedded across clinics who help facilitate adoption, provide hands-on support, and organize communication campaigns or events to raise awareness of the value and appropriate use of AI solutions.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Currently, hospitals using electronic health record (EHR) systems have fairly good integration of clinical data. However, there is a significant gap in integration of clinical data between organizations not using the same EHR system. This has significant safety implications when patients transfer between organizations. It also has a significant impact on the ability to screen patients for clinical trial options if their health records are stored in a different EHR system. AI is increasingly performative in finding the right patient for the right trial. However, without true clinical data integration, that is very difficult to automate.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

There is significant opportunity for AI to address patient education and monitoring in the home setting. Patients with life threatening or chronic diseases can find it hard to absorb information in the clinic setting. Providing direct to patient education tools that can be used and interactive in the home setting gives patients the ability to digest information in a way that is meaningful to them. Increased education and understanding of their disease can lead to better symptom monitoring and treatment plan adherence.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?
 - a. Are there published findings about the impact of adopted AI tools and their use clinical care?

- b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

Peer-reviewed publications are not consistent sources of organizations reporting outcomes and risks of AI solutions, especially failures of AI solution implementation. See question 2 on AI national registries.

Specific areas of research that can be considered for future funding are:

- *Automated evaluation methods for AI solutions (agentic AI, LLM, predictive AI)*
- *Scalable frameworks for AI management - infrastructure recommendations through clinical practice learnings*
- *New Robust AI methodologies*

Closing

Thank you for the opportunity to provide these comments. Duke Health's faculty, researchers, and physician scientists bring deep expertise across clinical care, health policy, and biomedical innovation, and have regularly served as trusted advisors to both the Executive Branch and Congress. We welcome continued engagement and stand ready to support HHS as it considers policies to accelerate the safe, effective, and responsible adoption of AI in healthcare.

Sincerely,

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